The Spectrum of an Operator

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May 3, 2026

Outline

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Revision

Definition

Let T be a linear operator, $\lambda \in \mathbb{C}$ is called an eigenvalue of T, if there exists $x \in H$ with $x \neq 0$ such that

$$Tx = \lambda x$$

Revision

Example

Consider

$$T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$
$$x \longmapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x$$

$1 \in \mathbb{C}$ *is an eigenvalue of* T, *since there is* $(1,0) \in \mathbb{R}^2$ *such that*

$$T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Revision

Theorem

Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \to \mathcal{H}$ then T is invertible $\iff \ker T^* = \{0\}$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in \mathcal{H}$.

Revision

Theorem

Let $\mathcal{H}$ *be a Hilbert space and* $T : \mathcal{H} \to \mathcal{H}$ *then* T *is invertible* $\iff \ker T^* = \{0\}$ *and* $\exists \alpha > 0$ *such that* $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in \mathcal{H}$.

Corollary

Let $\mathcal{H}$ *be a Hilbert space and* $T : \mathcal{H} \to \mathcal{H}$ *then* T *is invertible* $\iff \exists \beta > 0$ *such that* $\|T^*(x)\| \geqslant \beta \|x\|$ *and* $\exists \alpha > 0$ *such that* $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in \mathcal{H}$.

Revision

Proof.

If T is invertible, then so is T^*. So, by Theorem 1, $\exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha\|x\|$ for all $x \in H$. Conversely, if $\exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$, then if $T^*u = 0$ for some $u \in H$, then

$$0 = \|T^*u\| \geq \beta\|u\| \implies u = 0$$

So, $\ker T^* = \{0\}$. By Theorem 1, T is invertible. □

Revision

Definition

A linear operator $T : \mathcal{H} \to \mathcal{H}$ is said to be **normal** *if* $TT^* = T^*T$.

Definition

A linear operator $T : \mathcal{H} \to \mathcal{H}$ is said to be **self-adjoint** *if* $T = T^*$.

Definition

A linear operator $U : \mathcal{H} \to \mathcal{H}$ is said to be **uniatry** *if* $UU^* = U^*U = I$

Outline

Definition of Spectrum

Definition

Let T be a linear operator. The resolvent set of T, denoted by $\rho(T)$, is $\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is invertible}\}$. The spectrum of T, denoted by $\sigma(T)$ is $\sigma(T) = \mathbb{C} \backslash \rho(T)$, or equivalently, $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$

Definition of Spectrum

Remark

Notice that, if $T \in B(\mathcal{H})$, by Theorem 1, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \ker(T - \lambda I)^* = \{0\} \text{ and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \alpha > 0\}$$

Alternatively, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \|(T - \lambda I)^* x\| \geq \beta\|x\| \text{ and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \beta, \alpha > 0\}$$

by Corollary 1

Relationship between Eigenvalue and Spectrum

Theorem

If λ is an eigenvalue of T , then $\lambda \in \sigma(T)$

Relationship between Eigenvalue and Spectrum

Proof.

Suppose that $\lambda \notin \sigma(T)$, then $T - \lambda I$ is invertible. So, we have $(T - \lambda I)u = 0 \implies u = (T - \lambda I)^{-1}(T - \lambda I)u = (T - \lambda I)^{-1}(0) = 0$. So, if there is $u \in \mathcal{H}$ such that $Tu = \lambda u$, then $u = 0$. So, λ is not an eignvalue of T. $\square$

Relationship between Eigenvalue and Spectrum

Theorem

Let $\mathcal{H}$ be a finite dimensional Hilbert space, and let $T : \mathcal{H} \to \mathcal{H}$ be linear. If $\lambda \in \sigma(T)$, then λ is an eigenvalue of T.

Relationship between Eigenvalue and Spectrum

Remark

For finite dimesonial Hilbert space, the spectrum of T consist only the eigenvalues of T. However, in infinite dimesional Hilbert space, we may have no eigenvalues. For instance, consider the shift operator. Here, it has no eigenvalues, since if we consider $(T - \lambda I)x = 0$, we see that

$$(-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \cdots) = (0, 0, 0, \cdots)$$
$$\implies x = (0, 0, 0, \cdots)$$

So, T does not have an eigenvalue. However, we showed that T is not invertible. So, $T - 0I$ is not invertible and $0 \in \sigma(T)$. So, the spectrum of an operator does not only consist of eigenvalues.

Relationship between Eigenvalue and Spectrum

Proof.

Suppose that λ is not an eigenvalue of T, then there is nonzero $u \in \mathcal{H}$ such that $Tu = \lambda u$. So, there is no nonzero $u \in \mathcal{H}$ such that $(T - \lambda I)u = 0$. So, $\ker(T - \lambda I) = \{0\}$. By rank-nullity theorem, we have

$$\begin{aligned} \dim \mathcal{H} &= \dim \ker(T - \lambda I) + \dim \operatorname{Im}(T - \lambda I) \\ &= 0 + \dim \operatorname{Im}(T - \lambda I) \\ &= \dim \operatorname{Im}(T - \lambda I) \end{aligned}$$

So, $\operatorname{Im}(T - \lambda I) = \mathcal{H}$. This means that $(T - \lambda I)$ is surjective, and therefore, invertible. So, $\lambda \in \rho(T)$, and therefore, $\lambda \notin \sigma(T)$. □

Properties and Theorems

Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \to \mathcal{H}$ be linear operator. The following properties hold:

1. $\sigma(I) = \{1\}$

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2. $\sigma\left(T^*\right) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
3. *If T is invertible, then $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$*

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3. *If T is invertible, then $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$*
4. *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*

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Proposition

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2. $\sigma\left(T^*\right) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
3. *If T is invertible, then $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$*
4. *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*
5. *If $U : \mathcal{H} \to \mathcal{H}$ is invertible, then $\sigma(UTU^{-1}) = \sigma(T)$*

Properties and Theorems

Proof.

- **Proof of 1:** Suppose that $\lambda \in \sigma(I)$, then $I - \lambda I = (1 - \lambda)I$ is not invertible. $(1 - \lambda)I$ is not invertible if and only if $1 - \lambda = 0$, since $1 - \lambda \in \mathbb{F}$. So, this is true if and only if $\lambda = 1$.

Properties and Theorems

Proof.

- **Proof of 1:** Suppose that $\lambda \in \sigma(I)$, then $I - \lambda I = (1-\lambda)I$ is not invertible. $(1-\lambda)I$ is not invertible if and only if $1-\lambda = 0$, since $1 - \lambda \in \mathbb{F}$. So, this is true if and only if $\lambda = 1$.
- **Proof of 2:** Suppose that $\mu \in \sigma(T^*)$, then $T^* - \mu I$ is not invertible. Observe that $T^* - \mu I$ is invertible if and only if $(T^* - \mu I)^* = T - \bar{\mu} I$ is invertible, since if $(T^* - \mu I)^*$ is invertible, then $((T^* - \mu I)^*)^* = T^* - \mu I$ is invertible. So, $T^* - \mu I$ is not invertible iff $(T^* - \mu I)^* = T - \bar{\mu} I$ is not invertible. So, $\bar{\mu} \in \sigma(T)$, i.e. $\mu \in \{\bar{\lambda} : \lambda \in \sigma(T)\}$.

$\square$

Proof.

Properties and Theorems

Proof.

- **Proof of 3:** If T is invertible, i.e. $T - 0I$ is invertible, then $0 \notin \sigma(T)$. Similarly, $0 \notin \sigma(T^{-1})$. Suppose that $\mu \in \sigma(T^{-1})$, then $T^{-1} - \mu I$ is not invertible. Observe that $T^{-1} - \mu I$ is invertible if and only if $T(T^{-1} - \mu I) = I - \mu T$ is invertible, since if $T(T^{-1} - \mu I)$ has an inverse, K, then KT would be an inverse of $T^{-1} - \mu I$. So, $T^{-1} - \mu I$ is not invertible iff $T(T^{-1} - \mu I) = I - \mu T$ is not invertible. Also, $I - \mu T$ is not invertible iff $T - \mu^{-1} I$, where μ^{-1} exists, since $\mu \neq 0$. So, $\mu^{-1} \in \sigma(T)$, i.e. $\mu \in \{\lambda^{-1} : \lambda \in \sigma(T)\}$.

Properties and Theorems

Proof.

- **Proof of 3:** If T is invertible, i.e. $T - 0I$ is invertible, then $0 \notin \sigma(T)$. Similarly, $0 \notin \sigma(T^{-1})$. Suppose that $\mu \in \sigma(T^{-1})$, then $T^{-1} - \mu I$ is not invertible. Observe that $T^{-1} - \mu I$ is invertible if and only if $T(T^{-1} - \mu I) = I - \mu T$ is invertible, since if $T(T^{-1} - \mu I)$ has an inverse, K, then KT would be an inverse of $T^{-1} - \mu I$. So, $T^{-1} - \mu I$ is not invertible iff $T(T^{-1} - \mu I) = I - \mu T$ is not invertible. Also, $I - \mu T$ is not invertible iff $T - \mu^{-1} I$, where μ^{-1} exists, since $\mu \neq 0$. So, $\mu^{-1} \in \sigma(T)$, i.e. $\mu \in \{\lambda^{-1} : \lambda \in \sigma(T)\}$.
- **Proof of 5:** Suppose that $\lambda \in \sigma(UTU^{-1})$, then $UTU^{-1} - \lambda I$ is not invertible. Observe that $UTU^{-1} - \lambda I = U(T - \lambda)U^{-1}$ is not invertible iff $T - \lambda I$ is not invertible. So, $\lambda \in \sigma(T)$.

$\square$

Properties and Theorems

Example

Quantum Physics We have seen that for an observable t, we associate a self-adjoint operator T. **The eigenvalues of the operator T represent the possible values of that observable t.**

Properties and Theorems

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$$H\psi = E\psi = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi.$$

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Quantum Physics We have seen that for an observable t, we associate a self-adjoint operator T. **The eigenvalues of the operator T represent the possible values of that observable t.** *For example, if we want to find the possible values of the energy, we need to find the eigenvalues of H. In other words, we solve*

$$H\psi = E\psi = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi.$$

This is the famous **Schrödinger** *equation.*

Properties and Theorems

Theorem

Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$. If $|\lambda| > \|T\|$ then $\lambda \notin \sigma(T)$.

Properties and Theorems

Proof.

Let $u \in \ker(T - \lambda I)^*$, then $(T - \lambda I)^* u = (T^* - \bar{\lambda} I)u = 0$. So, $T^*u - \bar{\lambda}u = 0 \implies T^*u = \bar{\lambda}u$. So,

$$|\lambda|\|u\| = |\bar{\lambda}|\|u\| = \|\bar{\lambda}u\| = \|T^*u\| \leq \|T^*\|\|u\| = \|T\|\|u\|$$

$$\implies (|\lambda| - \|T\|)\|u\| \leq 0$$

Since $|\lambda| > \|T\|$, then $\|u\| \leq 0 \implies \|u\| = 0 \implies u = 0$. Hence, $\ker(T - \lambda I)^* = \{0\}$. □

Proof.

We also have that

$$\begin{aligned}
\|(T-\lambda I)x\| = &\|Tx-\lambda x\| \geq \big|\|Tx\| - |\lambda|\|x\|\big| \\
&= \left|\frac{\|Tx\|}{\|x\|} - |\lambda|\right| \|x\| \\
&= \left(|\lambda| - \frac{\|Tx\|}{\|x\|}\right)\|x\| \\
&\geq (|\lambda| - \|T\|)\|x\| \\
&= \alpha\|x\|
\end{aligned}$$

where $\alpha := |\lambda| - \|T\|$ is positive, since $|\lambda| > \|T\|$. So, $\lambda \in \rho(T)$, by first equality in Remark 1. Hence, $\lambda \notin \sigma(T)$. $\square$

Properties and Theorems

Corollary

If $T \in B(\mathcal{H})$, then $\sigma(T)$ is bounded.

Properties and Theorems

Proof.

If $\lambda \in \sigma(T)$, then, by the contrapositive of Theorem 4, $|\lambda| \leq \|T\|$. So, $\lambda \in \overline{B(0, \|T\|)}$. Hence, $\sigma(T) \subseteq \overline{B(0, \|T\|)}$, i.e. $\sigma(T)$ is bounded. $\square$

Properties and Theorems

Corollary

If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Properties and Theorems

Proposition

If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Proof.

If $|\lambda| > 1 = \|U\|$, then $\lambda \notin \sigma(U)$. Moreover, if $|\lambda| < 1$, then

$$U - \lambda I = -\lambda U(U^{-1} - \frac{1}{\lambda} I)$$

is invertible, since $-\lambda U$ is invertible and $\left|\frac{1}{\lambda}\right| > 1 = \|U^{-1}\|$. So, $\lambda \notin \sigma(U)$. Therefore, $|\lambda| = 1$ if $\lambda \in \sigma(U)$. $\square$

Properties and Theorems

Theorem

If $T \in B(\mathcal{H})$, then $\sigma(T) \subseteq \mathbb{C}$ is a closed set.

Properties and Theorems

Proof.

We will show that $\rho(T) = \mathbb{C} \backslash \sigma(T)$ is open. Let $\lambda \in \rho(T)$. This means that, by second equality in Remark 1, there is $\alpha > 0$ such that

$$\|(T - \lambda I)u\| \geq \alpha \|u\| \implies \frac{\|(T - \lambda I)u\|}{\|u\|} \geq \alpha$$

Since the quantity $\frac{\|(T-\lambda I)u\|}{\|u\|}$ has a lower bound, its infimum exists. Similarly, by second equality in Remark 1, the infimum of the quantity $\frac{\|(T-\lambda I)^* u\|}{\|u\|}$ exists. $\square$

Properties and Theorems

Proof.

Now, let

$$\gamma := \inf_{u \in H \setminus \{0\}} \frac{\|(T - \lambda I)u\|}{\|u\|} \quad \gamma' := \inf_{u \in G \setminus \{0\}} \frac{\|(T - \lambda I)^* u\|}{\|u\|}$$

and let $\delta := \frac{1}{2}\min(\gamma, \gamma')$. We want to show that

$$B(\lambda, \delta) \subseteq \rho(T)$$

$\square$

Properties and Theorems

Proof.

Let $\tilde{\lambda} \in B(\lambda, \delta)$, then $|\tilde{\lambda} - \lambda| < \delta$. First, we observe that

$$
\begin{aligned}
\|(T - \tilde{\lambda} I)u\| &= \|(T - \lambda I)u - (\tilde{\lambda} - \lambda)I(u)\| \\
&\geq \big| \|(T - \lambda I)u\| - |\bar{\lambda} - \lambda| \|u\| \big| \\
&= \left| \frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right| \|u\| \\
&= \left(\frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right) \|u\| \\
&> (\gamma - \frac{1}{2} \min(\gamma, \gamma')) \|u\| \\
&\geq (\gamma - \frac{\gamma}{2}) \|u\| \\
&= \frac{\gamma}{2} \|u\|
\end{aligned}
$$

Properties and Theorems

Proof.

Similarly, we have

$$
\begin{aligned}
\|(T-\tilde{\lambda} I)^* u\| &= \|(T-\lambda I)^* u - \overline{(\tilde{\lambda}-\lambda)} I(u)\| \\
&\geq \big|\|(T-\lambda I)^* u\| - |\overline{\tilde{\lambda}-\lambda}|\|u\|\big| \\
&= \big|\|(T-\lambda I)^* u\| - |\tilde{\lambda}-\lambda|\|u\|\big| \\
&= \left|\frac{\|(T-\lambda I)^* u\|}{\|u\|} - |\tilde{\lambda}-\lambda|\right|\|u\| \\
&= \left(\frac{\|(T-\lambda I)^* u\|}{\|u\|} - |\tilde{\lambda}-\lambda|\right)\|u\| \\
&> (\gamma' - \frac{1}{2}\min(\gamma,\gamma'))\|u\| \geq (\gamma' - \frac{\gamma'}{2})\|u\| = \frac{\gamma'}{2}\|u\|
\end{aligned}
$$

So, by the second equality in Remark 1, $\tilde{\lambda} \in \rho(T)$. Therefore, $B(\lambda,\delta) \subseteq \rho(T)$. □

Properties and Theorems

Corollary

If $T \in B(\mathcal{H})$, *then* $\sigma(T)$ *is compact in* $\mathbb{C}$

Outline

The Shift Operator

Example

We would like to find the spectrum of the unilateral shift, $T : \ell^2 \to \ell^2$*, defined by*

$$T(x_1, x_2, \cdots) = (0, x_1, x_2, \cdots)$$

Since T *has been shown to have no eigenvalue, we will consider the eigenvalues of* T^* *instead and use the properties of the spectrum to have a lower bound for the spectrum.*

The Shift Operator

Example

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Claim: $x \in \ell^2$

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Suppose that $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then consider the sequence

$$x = (1, \lambda, \lambda^2, \cdots)$$

Claim: $x \in \ell^2$

We have

$$\sum_{n=0}^{\infty} |x_n|^2 = \sum_{n=0}^{\infty} (|\lambda|^n)^2 = \sum_{n=0}^{\infty} (|\lambda|^2)^n = \frac{1}{1 - |\lambda|^2} < \infty$$

The Shift Operator

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Now, observe that

$$\begin{aligned} T^*x &= T^*(1, \lambda, \lambda^2, \cdots) \\ &= (\lambda, \lambda^2, \lambda^3, \cdots) = \lambda(1, \lambda, \lambda^2, \cdots) = \lambda x \end{aligned}$$

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Now, observe that

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So, every $\lambda \in \mathbb{C}$ *such that* $|\lambda| < 1$ *is an eigenvalue of* T^*.

The Shift Operator

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Now, observe that

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So, every $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$ is an eigenvalue of T^*. Now, we have

$$\{\lambda : |\lambda| < 1\} = \{\bar{\lambda} : |\lambda| < 1\} \subseteq \sigma(T)$$

The Shift Operator

Example

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

The Shift Operator

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So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0,1) \subseteq \sigma(T) \subseteq \overline{B(0,1)}$$

The Shift Operator

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So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0,1) \subseteq \sigma(T) \subseteq \overline{B(0,1)}$$

Since $\sigma(T)$ *is closed, we have*

$$\sigma(T) = \overline{B(0,1)} = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$